

# Algorithmic Impact Assessment Results

Version: 0.10.0

## Project Details

**1. Name of Respondent**

Derek Cotton

**2. Job Title**

Senior Business Analyst

**3. Department**

Employment and Social Development (Department of)

**4. Branch**

Integrity Services Branch

**5. Project Title**

Internal Activity and Access Monitoring

**6. Project ID from IT Plan**

201800028

**7. Project Phase**

Implementation

[Points: +0]

**8. Project Description**

This project will reduce risk for the department by acquiring and implementing an enterprise-wide solution to ensure compliance with audit recommendations to enhance investigation capabilities and oversight of ESDC information accessed and used by employees in core ESDC applications. The solution will reduce the time required to gather evidence, improve workload management by enabling automated dissemination of alerts and will enable automated detection of possible wrongdoing.

## About the System

**9. Please check which of the following capabilities apply to your system.**

Other (please specify)

[Points: +0]

**10. Other (please specify)**

As noted previously, the solution is specifically configured for the department to monitor application logs for a specific set of use cases of possible wrongdoing and misuse of information. It is also configured to only ingest log information from specific departmental applications that are in scope of the project. The solution in its current configuration, does not provide recommendations, does not make decisions, nor does it identify anomalies or make predictions.

## Section 1: Impact Level: 1

*Current Score: 30*

*Raw Score: 30*

*Mitigation Score: 31*

# Section 2: Requirements Specific to Impact Level 1

## Peer review

None

## Gender-based Analysis Plus

None

## Notice

None

## Human-in-the-loop-for-decisions

Decisions may be rendered without direct human involvement.

## Explanation

In addition to any applicable legal requirement, ensure that a meaningful explanation is published for common decision results. The explanation must provide a general description of:

- the role of the system in the decision-making process;
- input data, its source and method of collection;
- the criteria used to evaluate input data and the operations applied to process it;
- the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision; and
- the principal factors behind a decision.

Explanations must also inform clients of relevant recourse options, where appropriate.

Descriptions must be made available in plain language through the Algorithmic Impact Assessment and discoverable via a departmental website.

## Training

None

## IT and business continuity management

None

## Approval for the system to operate

None

# Other requirements

The Directive on Automated Decision-Making also includes other requirements that must be met for all impact levels.

[tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592](https://tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592)

Contact your institution's ATIP office to discuss the requirement for a Privacy Impact Assessment as per the Directive on Privacy Impact Assessment.

## Section 3: Questions and Answers

### 3.1: Impact Questions and Answers

#### 3.1.1: Reasons for Automation

**1. What is motivating your team to introduce automation into this decision-making process?**

Other (please specify) [Points: +0]

**2. What client needs will the system address and how will this system meet them? If possible, describe how client needs have been identified.**

The IAAM solution will provide near real-time capabilities to actively monitor the departmental core systems on an automated basis and will provide detailed and accurate evidence to support integrity investigative activities.

**3. Please describe any public benefits the system is expected to have.**

The benefits to the public are to ensure the integrity and security of personal information of Canadian citizens contained in ESDC systems and applications.

**4. How effective will the system likely be in meeting client needs?**

Very effective [Points: +0]

**5. Please describe any improvements, benefits, or advantages you expect from using an automated system. This could include relevant program indicators and performance targets.**

The new solution will provide complete and traceable evidence that will enhance investigative capabilities, analysis and integrity compliance/enforcement capabilities of the department. The solution is expected to increase the rate of detection of privacy breaches, wrongdoing and misuse of information and will reduce the length of time required to gather evidence.

**6. Please describe how you will ensure that the system is confined to addressing the client needs identified above.**

The solution is an off-the-shelf commercial solution specifically designed for threat monitoring and has been specifically configured for the department to monitor unique use cases of wrongdoing and misuse of information. It is also configured to only ingest log information from specific departmental applications in scope of the project.

**7. Please describe any trade-offs between client interests and program objectives that you have considered during the design of the project.**

No trade-offs identified.

**8. Have alternative non-automated processes been considered?**

Yes [Points: +0]

**9. If non-automated processes were considered, why was automation identified as the preferred option?**

Since 2018 the monitoring of departmental logs for core applications has been done manually using internally available statistical software SAS. The current manual process identifies unauthorized access or internal wrongdoing weeks after it may have occurred. In comparison with the enterprise and automated approach (through the IAAM solution), the current in-house log monitoring approach is very slow to expand, does not support near real-time alerts, and does not provide any case management functionality to organize and triage alerts.

**10. What would be the consequence of not deploying the system?**

Service delivery cannot achieve performance targets, Service quality is not as high, Service cannot be delivered in a timely or efficient manner [Points: +2]

**3.1.2: Risk Profile**

**11. Is the project within an area of intense public scrutiny (e.g. because of privacy concerns) and/or frequent litigation?**

No [Points: +0]

**12. Are clients in this line of business particularly vulnerable?**

No [Points: +0]

**13. Are stakes of the decisions very high?**

No [Points: +0]

**14. Will this project have major impacts on staff, either in terms of their numbers or their roles?**

No [Points: +0]

**15. Will the use of the system create or exacerbate barriers for persons with disabilities?**

No [Points: +0]

**3.1.3: Project Authority**

**16. Will you require new policy authority for this project?**

No [Points: +0]

**3.1.4: About the Algorithm**

**17. The algorithm used will be a (trade) secret**

No [Points: +0]

**18. The algorithmic process will be difficult to interpret or to explain**

No [Points: +0]

**3.1.5: About the Decision**

**19. Please describe the decision(s) that will be automated.**

IAAM will not make decisions autonomously. The solution will ingest log information from specific departmental applications and will flag cases of potential wrongdoing in near real-time for integrity officers based on likely markers of wrongdoing. The investigative officer(s) will then use the alerts as a referral, will review and validate information received and, if deemed necessary, will conduct a thorough investigation following program procedures.

**20. Does the decision pertain to any of the categories below:**

Other (please specify) [Points: +1]

**21. Other (please specify)**

Integrity Compliance.

### **3.1.6: Impact Assessment**

**22. Which of the following best describes the type of automation you are planning?**

Partial automation (the system will contribute to administrative decision-making by supporting an officer through assessments, recommendations, intermediate decisions, or other outputs)

[Points: +2]

**23. Please describe the role of the system in the decision-making process.**

The solution will ingest log information from specific departmental applications and will flag cases of potential wrongdoing in near real-time for integrity officers based on likely markers of wrongdoing. The investigative officer(s) will then use the alerts as a referral, will review and validate information received and, if deemed necessary, will conduct a thorough investigation following program procedures.

**24. Will the system be making decisions or assessments that require judgement or discretion?**

No

[Points: +0]

**25. Please describe the criteria used to evaluate client data and the operations applied to process it.**

The IAAM solution does not evaluate client data. Client activity/behaviour in key benefit applications is monitored using audit logs captured by the relevant application in scope of the IAAM solution monitoring. The disclosure of criteria used for the purposes of integrity management is restricted due to the nature of the solution.

**26. Please describe the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision.**

IAAM will provide officers with alerts that contain information pertaining to use cases which the officer(s) will review, validate and then, if deemed necessary, will use to conduct a thorough investigation following program procedures.

**27. Will the system perform an assessment or other operation that would not otherwise be completed by a human?**

No

[Points: +0]

**28. Is the system used by a different part of the organization than the ones who developed it?**

No

[Points: +0]

**29. Are the impacts resulting from the decision reversible?**

Reversible

[Points: +1]

**30. How long will impacts from the decision last?**

Impacts are most likely to be brief

[Points: +1]

**31. Please describe why the impacts resulting from the decision are as per selected option above.**

The alerts generated by the IAAM solution will be reviewed by investigators and may be used as evidence in the course of an investigation. Any decisions made regarding an investigation will be made by a human.

**32. The impacts that the decision will have on the rights or freedoms of individuals will likely be:**

Little to no impact

[Points: +1]

**33. Please describe why the impacts resulting from the decision are as per selected option above.**

The solution will have minimal to no impacts as it will simply provide integrity officers with a list of flagged cases based on likely markers of wrongdoing. The officer(s) will review and validate information received and, if deemed necessary, will conduct a thorough investigation following program procedures which will eliminate the likelihood of automation bias.

**34. The impacts that the decision will have on the equality, dignity, privacy, and autonomy of individuals will likely be:**

Little to no impact [Points: +1]

**35. Please describe why the impacts resulting from the decision are as per selected option above.**

The solution will have minimal to no impacts as it will simply provide integrity officers with a list of flagged cases based on likely markers of wrongdoing. The officer(s) will review and validate information received and, if deemed necessary, will conduct a thorough investigation following program procedures which will eliminate the likelihood of automation bias.

**36. The impacts that the decision will have on the health and well-being of individuals will likely be:**

Little to no impact [Points: +1]

**37. Please describe why the impacts resulting from the decision are as per selected option above.**

IAAM will not make decisions autonomously. The solution will ingest log information from specific departmental applications and will flag cases of potential wrongdoing in near real-time for integrity officers based on likely markers of wrongdoing. The investigative officer(s) will then use the alerts as a referral, will review and validate information received and, if deemed necessary, will conduct a thorough investigation following program procedures.

**38. The impacts that the decision will have on the economic interests of individuals will likely be:**

Little to no impact [Points: +1]

**39. Please describe why the impacts resulting from the decision are as per selected option above.**

IAAM will not make decisions autonomously. The solution will ingest log information from specific departmental applications and will flag cases of potential wrongdoing in near real-time for integrity officers based on likely markers of wrongdoing. The investigative officer(s) will then use the alerts as a referral, will review and validate information received and, if deemed necessary, will conduct a thorough investigation following program procedures.

**40. The impacts that the decision will have on the ongoing sustainability of an environmental ecosystem, will likely be:**

Little to no impact [Points: +1]

**41. Please describe why the impacts resulting from the decision are as per selected option above.**

IAAM will have no impact on resources required for operations, nor will it impact the frequency of investigations.

**3.1.7: About the Data - Data Source**

**42. Will the Automated Decision System use personal information as input data?**

Yes [Points: +4]

**43. Have you verified that the use of personal information is limited to only what is**

**directly related to delivering a program or service?**

Yes

[Points: +0]

**44. Is the personal information of individuals being used in a decision-making process that directly affects those individuals?**

No

[Points: +0]

**45. Have you verified if the system is using personal information in a way that is consistent with: (a) the current Personal Information Banks (PIBs) and Privacy Impact Assessments (PIAs) of your programs or (b) planned or implemented modifications to the PIBs or PIAs that take new uses and processes into account?**

Yes

[Points: +0]

**46. Please list relevant PIB Bank Numbers.**

PSE 901 and PSU 905

**47. What is the highest security classification of the input data used by the system? (Select one)**

Protected B / Protected C

[Points: +3]

**48. Who controls the data?**

Federal government

[Points: +1]

**49. Will the system use data from multiple different sources?**

Yes

[Points: +4]

**50. Will the system require input data from an Internet- or telephony-connected device? (e.g. Internet of Things, sensor)**

No

[Points: +0]

**51. Will the system interface with other IT systems?**

Yes

[Points: +4]

**52. Who collected the data used for training the system?**

Your institution

[Points: +1]

**53. Who collected the input data used by the system?**

Your institution

[Points: +1]

**54. Please describe the input data collected and used by the system, its source, and method of collection.**

Data collected is ingested by the IAAM solution following a prescribed schedule and is collected from relevant ESDC application data sources in scope of the IAAM solution.

### **3.1.8: About the Data - Type of Data**

**55. Will the system require the analysis of unstructured data to render a recommendation or a decision?**

No

[Points: +0]

## **3.2: Mitigation Questions and Answers**

**3.2.1: Consultations - Have you engaged with any of the following groups?**

**1. Internal Stakeholders (federal institutions, including the federal public service)**

Yes

[Points: +1]

**2. Which Internal Stakeholders have you engaged?**

Office of the Privacy Commissioner of Canada, Access to Information and

|                                                                                                                                              |              |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Privacy Office, Legal Services, Client Experience / Client Relationship Management, Communications services, Program Policy, Data Governance | [Points: +0] |
| <b>3. External Stakeholders (groups in other sectors or jurisdictions)</b>                                                                   |              |
| Yes                                                                                                                                          | [Points: +1] |
| <b>4. Which External Stakeholders have you engaged?</b>                                                                                      |              |
| Other (please specify)                                                                                                                       | [Points: +0] |
| <b>5. Other (please specify)</b>                                                                                                             |              |
| Public Service Unions                                                                                                                        |              |

### 3.2.2: De-Risking and Mitigation Measures - Data Quality

|                                                                                                                                                                                                                         |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>6. Do you have documented processes in place to test datasets against biases and other unexpected outcomes? This could include experience in applying frameworks, methods, guidelines or other assessment tools.</b> |              |
| No                                                                                                                                                                                                                      | [Points: +0] |
| <b>7. Is this information publicly available?</b>                                                                                                                                                                       |              |
| No                                                                                                                                                                                                                      | [Points: +0] |
| <b>8. Have you developed a process to document how data quality issues were resolved during the design process?</b>                                                                                                     |              |
| Yes                                                                                                                                                                                                                     | [Points: +1] |
| <b>9. Is this information publicly available?</b>                                                                                                                                                                       |              |
| No                                                                                                                                                                                                                      | [Points: +0] |
| <b>10. Have you undertaken a Gender Based Analysis Plus of the data?</b>                                                                                                                                                |              |
| No                                                                                                                                                                                                                      | [Points: +0] |
| <b>11. Is this information publicly available?</b>                                                                                                                                                                      |              |
| No                                                                                                                                                                                                                      | [Points: +0] |
| <b>12. Have you assigned accountability in your institution for the design, development, maintenance, and improvement of the system?</b>                                                                                |              |
| Yes                                                                                                                                                                                                                     | [Points: +2] |
| <b>13. Do you have a documented process to manage the risk that outdated or unreliable data is used to make an automated decision?</b>                                                                                  |              |
| Yes                                                                                                                                                                                                                     | [Points: +2] |
| <b>14. Is this information publicly available?</b>                                                                                                                                                                      |              |
| No                                                                                                                                                                                                                      | [Points: +0] |
| <b>15. Is the data used for this system posted on the Open Government Portal?</b>                                                                                                                                       |              |
| No                                                                                                                                                                                                                      | [Points: +0] |

### 3.2.3: De-Risking and Mitigation Measures - Procedural Fairness

|                                                                                                                         |              |
|-------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>16. Does the audit trail identify the authority or delegated authority identified in legislation?</b>                |              |
| No                                                                                                                      | [Points: +0] |
| <b>17. Does the system provide an audit trail that records all the recommendations or decisions made by the system?</b> |              |
| Yes                                                                                                                     | [Points: +2] |
| <b>18. Are all key decision points identifiable in the audit trail?</b>                                                 |              |
| Yes                                                                                                                     | [Points: +2] |



19. Are all key decision points within the system's logic linked to the relevant legislation, policy or procedure?

Yes

[Points: +1]
20. Do you maintain a log detailing all of the changes made to the model and the system?

Yes

[Points: +2]
21. Does the system's audit trail clearly set out all decision points made by the system?

Yes

[Points: +1]
22. Can the audit trail generated by the system be used to help generate a notification of the decision (including a statement of reasons or other notification) where required?

No

[Points: +0]
23. Does the audit trail identify precisely which version of the system was used for each decision it supports?

No

[Points: +0]
24. Does the audit trail show who the authorized decision-maker is?

No

[Points: +0]
25. Is the system able to produce reasons for its decisions or recommendations when required?

Yes

[Points: +2]
26. Is there a process in place to grant, monitor, and revoke access permission to the system?

Yes

[Points: +1]
27. Is there a mechanism to capture feedback by users of the system?

Yes

[Points: +1]
28. Is there a recourse process planned or established for clients that wish to challenge the decision?

Yes

[Points: +2]
29. Does the system enable human override of system decisions?

Yes

[Points: +2]
30. Is there a process in place to log the instances when overrides were performed?

Yes

[Points: +1]
31. Does the system's audit trail include change control processes to record modifications to the system's operation or performance?

Yes

[Points: +2]
32. Have you prepared a concept case to the Government of Canada Enterprise Architecture Review Board?

Yes

[Points: +1]

**3.2.4: De-Risking and Mitigation Measures - Privacy**

33. If your system uses or creates personal information, have you undertaken or have you undertaken a Privacy Impact Assessment, or updated an existing one?

Yes

[Points: +1]
34. Please indicate the following in your answer: Title and scope of the Privacy Impact Assessment; How the automation project fits into the program; and Date of Privacy Impact Assessment completion or modification.

Privacy Impact Assessment Report - Internal Activity and Access Monitoring Solution

Phase 1: Employment Insurance (EI) Applications. The PIA was completed in July 2023.

There is currently a PIA being developed for Phase 2 of the project (to onboard SIN/SIR).

The IAAM solution will automate existing manual analytical processes and will provide near real-time monitoring as well as comprehensive digital evidence to investigators for significantly better outcomes in the prevention and detection of unauthorized access, use and insider threats and vulnerabilities to personal information. Accordingly, the solution will rely on the use of personal information held in various ESDC databases and systems.

This is a multi-year, multi-phase implementation led by Integrity Services Branch (ISB) starting with the implementation of IAAM on Employment Insurance (EI) systems, to be followed by the Social Insurance Number/Social Insurance Registry (SIN/SIR) business line.

**35. Have you design and build security and privacy into your systems from the concept stage of the project?**

Yes [Points: +1]

**36. Is the information used within a closed system (i.e. no connections to the Internet, Intranet or any other system)?**

No [Points: +0]

**37. If the sharing of personal information is involved, has an agreement or arrangement with appropriate safeguards been established?**

Yes [Points: +1]

**38. Will you de-identify any personal information used or created by the system at any point in the lifecycle?**

Yes [Points: +1]

**39. Please describe your de-identification method(s).**

Dashboards and reports produced by the solution will contain aggregated information and will not contain personal information.